

The Model-View-Controller (MVC) design pattern divides an application into three interconnected components:

- Model: Manages the data and business logic.
- View: Handles the presentation layer and user interface.
- Controller: Processes user input and interacts with the Model and View.

This separation facilitates modular development, easier testing, and maintenance.

Variants of MVC

① Model-View-Presenter (MVP):

- Presenter: Acts as an intermediary between the view and Model handling all presentation logic.
- View: Handles the presentation layer and user interface.
- Controller: Processes user input and interacts with the Model and View.

This separation facilitates modular development, easier testing and maintenance.

Variants of MVC

① Model-View-Presenter (MVP):

- Presenter: Acts as an intermediary between the View and Model handling all presentation logic.
- Model-View-ViewModel (MVVM):
 - ViewModel: Exposes data and commands from the Model to the View, facilitating data binding.
 - Use Case: Ideal for applications using data-binding capabilities, like WPF or Angular applications.